IE 206: Introduction to AI & ML

Introduction to Support Vector Machines(SVM)

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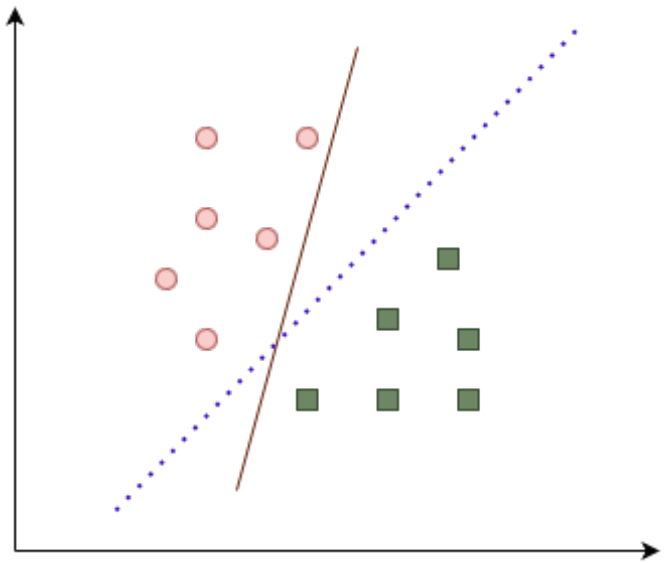
Linear Separability

- $S = \{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$ is a set of data points
- $x_i \in \mathcal{X}$ for all $i = 1, 2, \ldots, m$ are drawn iid
- $y_i \in \mathcal{Y}$ is the label of x_i.
- The data maybe linearly separable, i.e. there exists halfspace (w, b) such that $y_i = sign(\langle w, x_i \rangle + b)$.

Margin

Suppose we have a linearly separable data



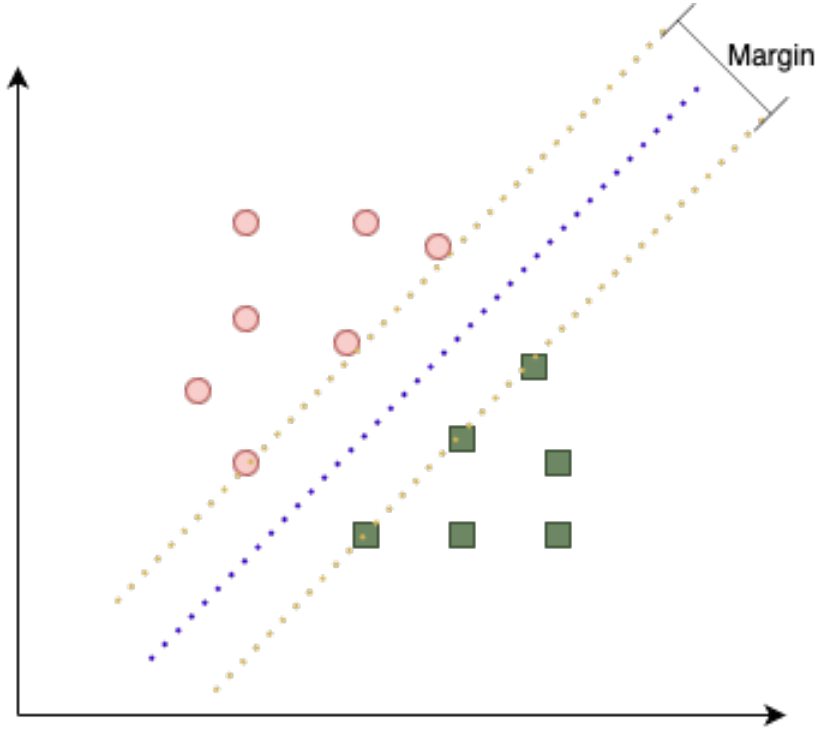
Which line would one choose, the dotted line or the brown line?

Definition

The margin of a hyperplane with respect to a training set is defined to be the minimal distance between a point in the training set and the hyperplane.

Margin

- The distance between a point x and the hyperplane defined by (w, b) where $\|w\| = 1$ is $|\langle w, x_i \rangle + b|$.



Hard-SVM

Claim: Consider a hypothesis (w, b) such that $\| w \| = 1$. Then the smallest distance of any point x from the hyperplane is $|\langle w, x \rangle + b|$.

Smallest Distance from x:

$$\min_{v : \langle w, v \rangle + b = 0} \| x - v \|$$

Margin:

$$\left(\min_{i \in [m]} |\langle w, x_i \rangle + b| \right)$$

Hard-SVM Rule:

- Hard-SVM is the learning rule in which we return an hyperplane that separates the training set with the largest possible margin given a dataset S.
- Hard-SVM rule stresses on classifying all data points correctly.(Not possible if data is not linearly separable.)

$$\max_{(w,b)} \left(\min_{i \in [m]} |\langle w, x_i \rangle + b| \right)$$
$$\text{such that} \quad y_i(\langle w, x_i \rangle + b) > 0 \;\; \forall \;\; i$$

Denote the optimal values is (w^*, b^*)

Hard-SVM

Whenever the data is linearly separable, we have an equivalent formulation:

$$(w^*, b^*) = \underset{(w,b)}{\operatorname{argmax}} \left(\min_{i \in [m]} y_i(\langle w, x_i \rangle + b) \right)$$

Steps:

- Define $\mathcal{G} = \{(w, b) : y_i(\langle w, x_i \rangle + b) > 0 \ \ \forall i\}$
- Show that

$$\underset{(w,b)}{\operatorname{argmax}} \left(\min_{i \in [m]} y_i(\langle w, x_i \rangle + b) \right) \in \mathcal{G}$$

- Show that $\forall \ (w, b) \in \mathcal{G}$,

$$\min_{i \in [m]} y_i(\langle w, x_i \rangle + b) = \min_{i \in [m]} |(\langle w, x_i \rangle + b)|$$

Hard-SVM Algorithm

An equivalent formulation of the Hard-SVM rule as a quadratic optimization problem is the following:

Hard SVM Algorithm

Input: $S = \{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$

$$(w_0, b_0) = \underset{(w,b)}{\operatorname{argmax}} \|w\|^2 \tag{1}$$

$$\text{such that } \; y_i(\langle w, x_i \rangle + b) \geq 1 \;\; \forall i \tag{2}$$

Output: $\hat{w} = \frac{w_0}{\|w_0\|}, \hat{b} = \frac{b_0}{\|w_0\|}$

Hard-SVM Lemma

Lemma: The output of Hard-SVM Algorithm is a solution of the Hard-SVM Rule.

Proof:

- Let (w^*, b^*) be a solution of Hard-SVM Rule and define the margin achieved by (w^*, b^*) to be

$$\gamma^* = \min_{i \epsilon [m]} y_i(\langle w^*, x_i \rangle + b^*)$$

- Therefore, $\forall\ i$ we have $y_i(\langle w^*, x_i \rangle + b^*) \geq \gamma^*$ or equivalently

$$y_i(\langle \frac{w^*}{\gamma^*}, x_i \rangle + \frac{b^*}{\gamma^*}) \geq 1$$

- The pair $(\frac{w^*}{\gamma^*}, \frac{b^*}{\gamma^*})$ satisfies the conditions of the quadratic optimization in the Hard-SVM Algorithm.

Therefore, $\|w_0\| \leq \|\frac{w^*}{\gamma^*}\| = \frac{1}{\gamma^*}$

Hard-SVM Lemma

- It follows that for all i, $y_i(\langle \hat{w}, x_i \rangle + \hat{b}) = \frac{1}{\|w_0\|} y_i(\langle \hat{w}, x_i \rangle + \hat{b}) \geq \frac{1}{\|w_0\|} \geq \gamma^*$
- Since $\|\hat{w}\| = 1$ we obtain that $(\hat{w}, \hat{b})$ is an optimal solution of the Hard-SVM Rule.

Homogenous case

The Homogeneous Case:

- Halfspaces that pass through the origin.
- It is often more convenient to consider homogenous halfspaces.

Set the bias term b zero and define label as $sign(<w, x>)$. For homogenous halfspaces Hard-SVM Rule is equivalent to:

$$w^* = \underset{w}{\operatorname{argmax}} \|w\|^2 \qquad \text{such that} \quad y_i(\langle w, x_i \rangle) \geq 1 \;\; \forall i \tag{3}$$

Soft SVM

- Hard SVM assumes linear separability of data which is a strong assumption
- Fore relaxing this strong assumption, we introduce slack variables $\xi_1, \xi_2 .... \xi_m$ such that $y_i(\langle w, x_i \rangle + b) \geq 1 - \xi_i$ where ξ_i measures how much we can violate the hard svm condition.
- Soft SVM jointly optimise the margin and average of ξ_i which controls the violation. This tradeoff is controlled by λ

Soft SVM

Soft SVM Algorithm

input: $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m)$
parameter: $\lambda > 0$
solve:

$$\min_{\mathbf{w}, b, \boldsymbol{\xi}} \left(\lambda \|\mathbf{w}\|^2 + \frac{1}{m} \sum_{i=1}^{m} \xi_i \right)$$

$$\text{s.t. } \forall i, \quad y_i \big(\langle \mathbf{w}, \mathbf{x}_i \rangle + b \big) \geq 1 - \xi_i \quad \text{and} \quad \xi_i \geq 0$$

output: $\mathbf{w}, b$

Hard SVM vs Soft SVM

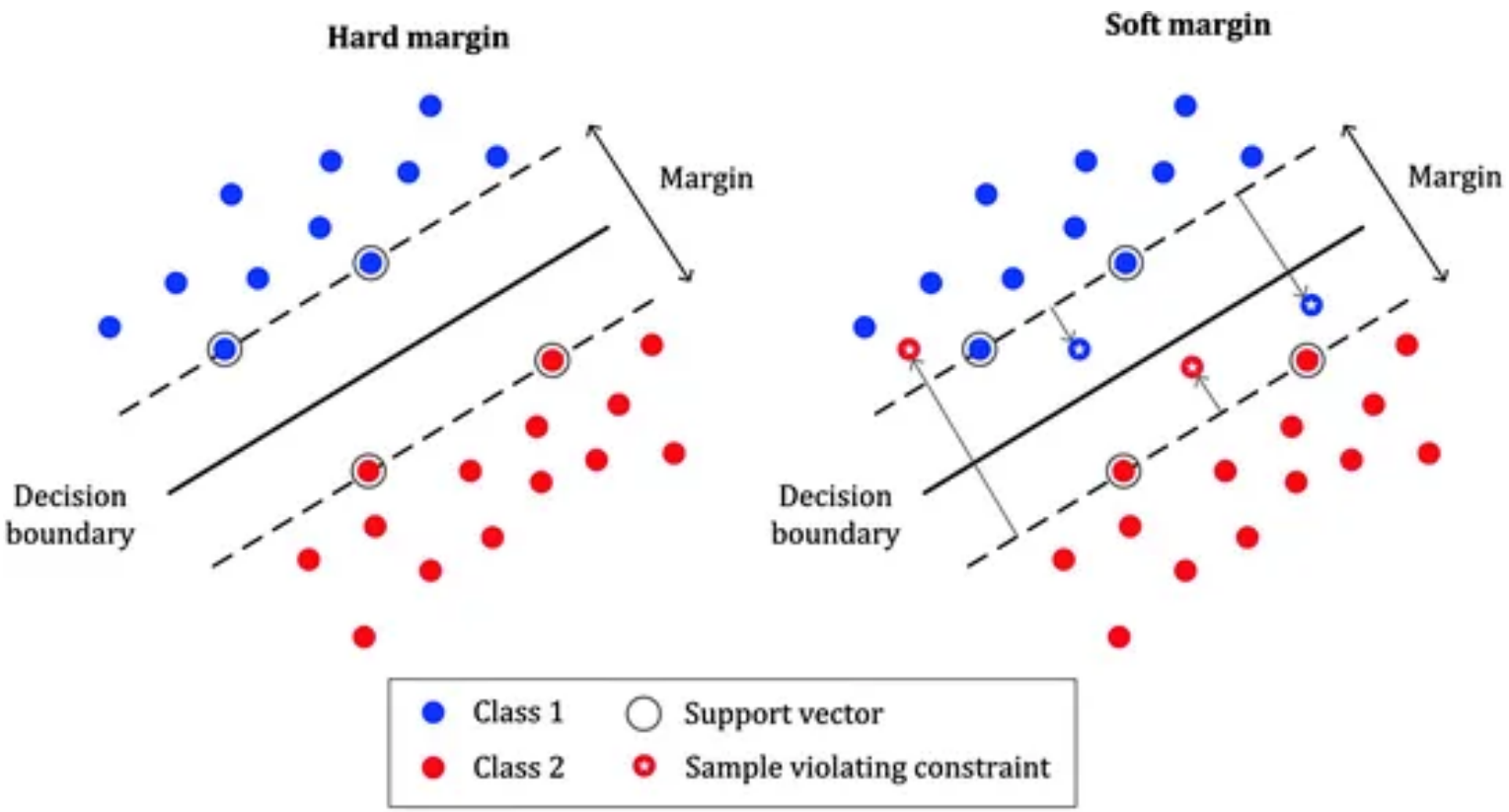


Figure: Hard SVM vs Soft SVM

[MathML.io]

Norm Regularisation

- We define hinge loss as

$$\ell_{\text{hinge}}((\mathbf{w}, b), (\mathbf{x}, y)) = \max\{0, 1 - y(\langle \mathbf{w}, \mathbf{x} \rangle + b)\}.$$

- The soft svm optimisation problem can now be taken as

$$\min_{\mathbf{w}, b} \lambda \|\mathbf{w}\|_2 + \mathcal{L}_{\text{hinge}}((\mathbf{w}, b)).$$

where $\mathcal{L}_{hinge}$ is average hinge loss over the dataset S and λ is a regularisation parameter

Optimality condition of Support Vectors

The name "Support Vector Machine" stems from the fact that the solution of Hard-SVM, w_0, is supported by (i.e., is in the linear span of) the examples that are exactly at a distance $\frac{1}{||w_0||}$ from the separating hyperplane. These vectors are therefore called support vectors. To see this, we rely on Fritz John optimality lemma.

Fritz John Lemma

Suppose that $w^* \epsilon \arg\min_w f(w)$ s.t. $\forall i \in [m], g_i(w) \leq 0$ where $f, g_1, \ldots, g_m$ be differentiable functions. Then, there exists $\alpha \in \mathbb{R}^m$ such that:

$$\nabla f(w^\star) + \sum_{i \in I} \alpha_i \nabla g_i(w^\star) = 0,$$

where

$$I = \{i : g_i(w^\star) = 0\}.$$

Optimality condition of Support Vectors

Theorem

Let w_0 be as defined in the homogeneous case(Hard-SVM) and let $I = \{i : |\langle w_0, x_i \rangle| = 1\}$. Then, there exist coefficients $\alpha_1, \ldots, \alpha_m$ such that

$$w_0 = \sum_{i \in I} \alpha_i x_i.$$

The examples $\{x_i : i \in I\}$ are called support vectors.
The proof of this theorem follows by applying the above lemma to the homogeneous case of Hard-SVM Rule.

Duality conditions

- Hard svm minimisation equation

$$\min_{\mathbf{w}} \|\mathbf{w}\|_2^2 \quad \text{subject to} \quad \forall i,\ y_i \langle \mathbf{w}, \mathbf{x}_i \rangle \geq 1.$$

  This equation can be rewritten as follows

$$g(\mathbf{w}) = \max_{\alpha \in \mathbb{R}^m,\, \alpha \geq 0} \sum_{i=1}^{m} \alpha_i \left(1 - y_i \langle \mathbf{w}, \mathbf{x}_i \rangle\right)$$

$$\text{subject to:} \quad \begin{cases} \text{If } \forall i, y_i \langle \mathbf{w}, \mathbf{x}_i \rangle \geq 1, & g(\mathbf{w}) = 0, \\ \text{Otherwise,} & g(\mathbf{w}) = \infty. \end{cases}$$

- Then the equation can be modified as

$$\min_{\mathbf{w}} (\|\mathbf{w}\|_2^2 + g(\mathbf{w}))$$

Duality conditions

$$\min_{\mathbf{w}} \max_{\alpha \in \mathbb{R}^m,\, \alpha \geq 0} \left(\frac{1}{2}\|\mathbf{w}\|_2^2 + \sum_{i=1}^{m} \alpha_i \left(1 - y_i \langle \mathbf{w}, \mathbf{x}_i \rangle\right) \right)$$

$$\geq \max_{\alpha \in \mathbb{R}^m,\, \alpha \geq 0} \min_{\mathbf{w}} \left(\frac{1}{2}\|\mathbf{w}\|_2^2 + \sum_{i=1}^{m} \alpha_i \left(1 - y_i \langle \mathbf{w}, \mathbf{x}_i \rangle\right) \right)$$

This inequality is called weak duality. In some cases, this inequality becomes equality and then it is called strong duality

Dual optimization Problem for SVM

- The problem with respect to $\mathbf{w}$ is unconstrained and the objective is differentiable; thus, at the optimum, the gradient equals zero:

$$\mathbf{w} - \sum_{i=1}^{m} \alpha_i y_i x_i = 0 \quad \Rightarrow \quad \mathbf{w} = \sum_{i=1}^{m} \alpha_i y_i x_i.$$

- This shows that the solution must be in a linear span of the examples. The optimization problem is given by:

$$\max_{\alpha \in \mathbb{R}^m : \alpha \geq 0} \left(\frac{1}{2} \left\| \sum_{i=1}^{m} \alpha_i y_i x_i \right\|^2 + \sum_{i=1}^{m} \alpha_i \left(1 - y_i \left\langle \sum_{j} \alpha_j y_j x_j, x_i \right\rangle \right) \right)$$

Dual optimization problem for SVM

- Rearranging yields the dual problem:

$$\max_{\alpha \in \mathbb{R}^m : \alpha \geq 0} \left(\sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y_i y_j \langle x_j, x_i \rangle \right)$$

- Since dual problem only involves inner products between instances and does not require direct access to specific elements within an instance

Summary

- SVMs can be used to learn half-spaces
- There are two types- hard and soft margin SVM
- Hard SVM seperates data perfectly with largest margin
- Soft SVM allows constraints to be violated for some extend